

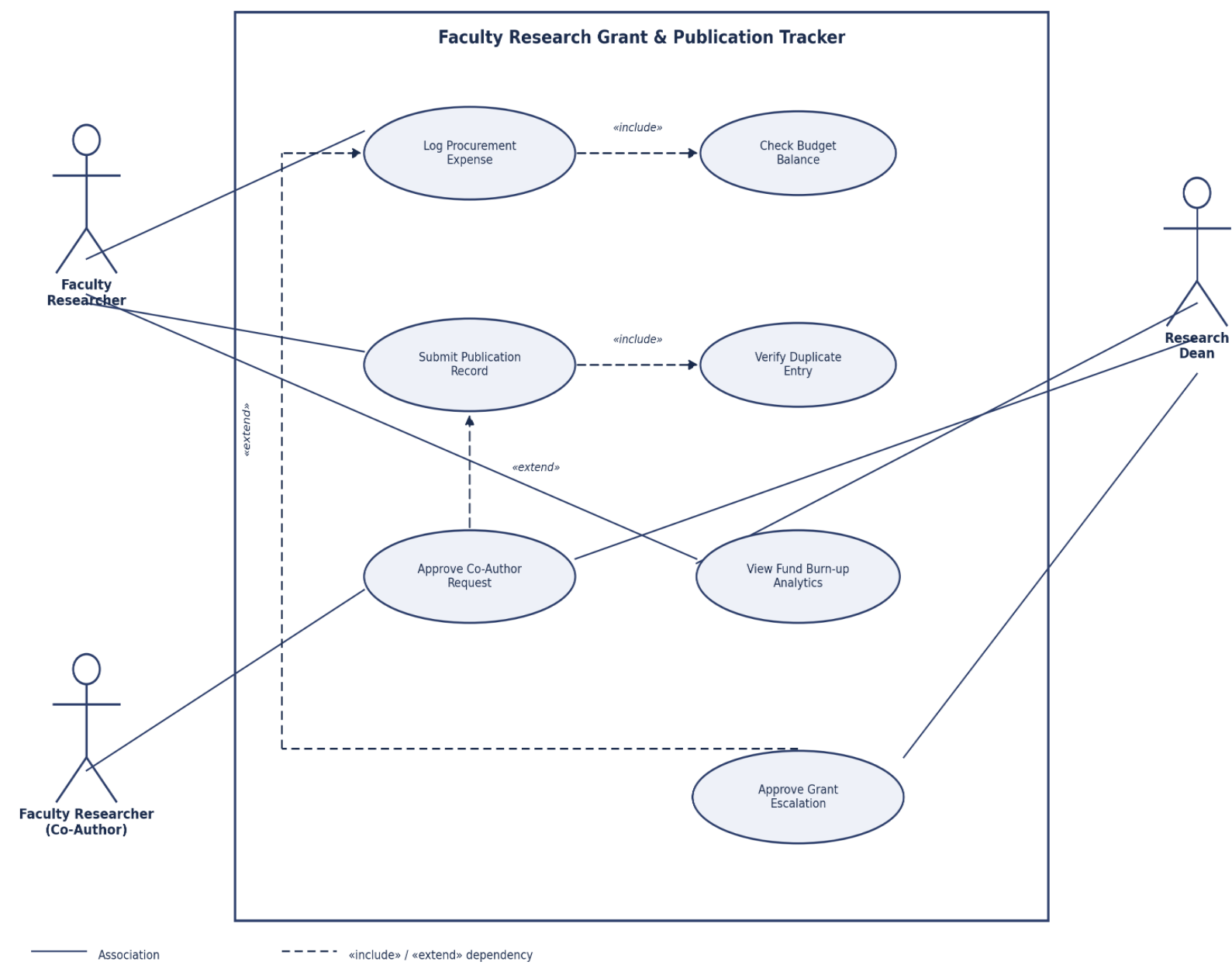
Faculty Research Grant & Publication Tracker

Complete Requirements Table — Problem Statement #09

Functional Requirements					
ID	Description		Priority	Acceptance Criteria	Rationale
FR-001	The system shall track grant fund allocations, allow faculty to log equipment/consumable procurement expenses against a grant, and compute the remaining budgetary balance in real time.		High	Pass: Balance updates accurately after each expense and the system blocks any expense that would exceed the grant cap unless an admin override is applied. Fail: Overdraft is allowed without an admin override, or the displayed balance does not match the sum of allocations minus expenses.	Prevents grant overdrafts and gives faculty real-time visibility into spendable funds, which is the core financial control the research deanery needs.
FR-002	The system shall allow faculty to submit publication records including title, authors, venue, DOI, indexing status (Scopus/WoS/UGC-CARE), and citation count, and shall check each submission for duplicate DOI/title entries before acceptance.		High	Pass: A duplicate DOI or high-similarity title is detected and the submission is blocked with a clear conflict message; unique submissions are stored successfully. Fail: A duplicate record is silently created, or a valid unique submission is rejected.	Accurate, de-duplicated publication data is essential for correct citation/indexing metrics reported to the deanery and external ranking bodies.
FR-003	For publications with more than one author, the system shall route a co-author approval request to each declared co-author and shall only mark the publication as “verified” once every co-author has approved the authorship order and contribution share.		Medium	Pass: Publication status changes to “verified” only after 100% of co-authors approve; a single rejection reverts the record to draft. Fail: A publication is marked verified while a co-author approval is still pending or rejected.	Protects academic integrity and ensures all contributing authors have consented to how credit is recorded before it counts toward departmental metrics.
FR-004	The system shall generate fund burn-up analytics, visually plotting cumulative grant expenditure against sanctioned budget and time-based milestones, filterable by grant, category, and date range.		Medium	Pass: The burn-up chart renders correct cumulative totals matching the underlying ledger for any selected filter combination. Fail: Chart totals diverge from the ledger, or filters produce no data / incorrect date ranges.	Gives the Research Dean an at-a-glance view of spending velocity across grants, supporting proactive budget management ahead of grant closure deadlines.
FR-005	The system shall automatically notify the Research Dean whenever a grant's cumulative fund utilization crosses the 75%, 90%, or 100% threshold, and shall require Dean approval before further expenses can be logged past the breached threshold.		High	Pass: A notification is generated within the same transaction that crosses a threshold, and further expense logging is blocked until the Dean approves or rejects escalation. Fail: Expenses continue to be logged past a threshold with no notification or approval gate.	Provides an early-warning and control mechanism so grants are not exhausted or overdrawn without deanery visibility and sign-off.
Non-Functional Requirements					
ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Security & Auditability	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals (expense logging, escalation decisions) and journal/publication status modifications, recording actor, timestamp, and before/after state.	High	Pass: Benchmarking and penetration tests confirm every financial/status-change event produces a cryptographically verifiable ledger entry that cannot be edited or deleted, under simulated peak load. Fail: Any ledger entry can be silently altered, or an approval/status change occurs without a corresponding entry.	Sponsored research funds and academic records require a defensible, tamper-proof trail for internal audits and external funding-agency compliance reviews.

NFR-002	Availability & Scalability	The system shall be available with at least 99.5% uptime during active grant-submission and publication-reporting windows, and shall support at least 500 concurrent authenticated users without degradation of response time beyond 2 seconds for standard read operations.	Medium	Pass: Load testing at 500 concurrent users maintains sub-2-second response for standard reads and uptime monitoring shows ≥99.5% availability over a rolling 30-day window. Fail: Response times exceed target under load, or uptime falls below 99.5% during a reporting window.	The portal must remain responsive during peak periods such as grant-renewal deadlines and annual publication reporting cycles, when faculty and deanery usage spikes simultaneously.
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UML USE CASE DIAGRAM:



Use Case ID: UC-02

Use Case Name: Submit Publication Record

Primary Actor(s): Faculty Researcher

Secondary Actor(s): System (Indexing Validator)

Related FR(s): FR-002

Relationship: «include» Verify Duplicate Entry

Brief Description

Enables a Faculty Researcher to submit a publication (journal/conference paper) with its indexing status (Scopus, Web of Science, UGC-CARE) and citation metrics. Every submission mandatorily triggers a duplicate-check sub-process to preserve data integrity across the tracker.

Preconditions

- The Faculty Researcher is authenticated and associated with at least one active or closed grant, or is submitting an unsponsored publication.

- The publication has a valid DOI or manuscript identifier available for entry.

Postconditions

- A new publication record is created in “pending verification” status.
- The record is linked to the relevant grant (if sponsored) for burn-up and output reporting.
- An audit-ledger entry documents the submission event.

Main Success Scenario

1. Faculty Researcher selects “Submit Publication” and enters title, authors, venue, DOI, and indexing status.
2. Faculty Researcher optionally links the publication to a sponsored grant.
3. System <<includes>> Verify Duplicate Entry to check the DOI/title against existing records.
4. Verify Duplicate Entry confirms no existing match is found and returns control to the main flow.

5. System stores the publication record with status “pending verification” and timestamps the submission.
6. System notifies the Research Dean that a new publication is awaiting review.
7. System displays a submission confirmation to the Faculty Researcher.

Alternate Flow — Duplicate Publication Detected

1. At step 3, Verify Duplicate Entry finds an existing record with a matching DOI or a high title-similarity score.
2. System halts the submission and displays the conflicting record to the Faculty Researcher.
3. Faculty Researcher may amend the DOI/title if the match was a false positive and resubmit, resuming at step 1.
4. If the Faculty Researcher confirms it is a genuine duplicate, the use case terminates without creating a new record.

Exception / Error Flow

If the indexing status field is left blank, the system rejects the submission at step 1 with a validation message.

If the linked grant has been closed for more than 90 days, the system disallows linking and requires the publication to be submitted as unsponsored.

Notes / UML Relationship Detail

Verify Duplicate Entry is modeled with an «include» relationship because duplicate checking is compulsory, identical logic reused on every publication submission, and independently testable as its own scenario.